

Mohammad Sadegh Sirjani

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Education

Ph.D., Computer Science	Jan. 2025 - Present
University of Texas at San Antonio, GPA: 3.94/4.0	
B.Sc., Computer Engineering	Sep. 2018 - Feb. 2024
Ferdowsi University of Mashhad, Cumulative GPA: 3.02/4.0 (16.52/20), Last 60 Credits: 3.89/4.0	

Research Interests

Internet of Things	Tiny AI	Edge AI
Embedded Systems	Energy Harvesting	Intermittent Computing

Research Experience

Research Assistant	Jan. 2025 - Present
ASIC Lab, University of Texas at San Antonio, Advisor: Dr. Mimi Xie	
- Developing energy-efficient neural network accelerators for edge AI applications on IoT devices. - Investigating ultra-low-power ML inference using hardware-software co-design approaches.	

Research Assistant	Aug. 2022 - Jan. 2024
Software Quality Lab, Ferdowsi University of Mashhad, Advisor: Dr. Behshid Behkamal	
- Analyzed Java application execution traces using XRebel and JRebel for software enhancement. - Developed code analysis techniques to identify class, interface, and enum connections. - Built tool for recommending microservice migration strategies from monolithic codebases.	

Research Assistant	Dec. 2023 - Aug. 2024
Web Technology Lab, Ferdowsi University of Mashhad, Advisor: Dr. Behshid Behkamal	
- Designed database structure and web-based API for psychological survey data collection. - Integrated LLMs to generate enhanced client assessment reports from survey data. - Successfully deployed production application through iterative prompt engineering refinements.	

Publications

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- Ahmad, M., **Sirjani, M. S.**, Wei, W., & Xie, M. (2026). "MOSAIC: Multi-Domain Sensing Architecture for Efficient Computing in Domain Wall Memory". *Under review at the Design, Automation and Test in Europe Conference (DATE 2027)*.
 - Sirjani, M. S.**, Ahmad, M., Wei, W., Dutton, P., Yanez, D., Pan, C., & Xie, M. (2026). "SparseBeat: Adaptive Compressed Sensing for Arrhythmia Classification on Energy-Harvesting Wearables". *Under review at the ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (SenSys 2027)*.
 - Mousavi, A., Nourbakhsh, E., **Sirjani, M. S.**, Slavin, R., Xie, M., Neely, L., Davis, J., & Quarles, J. (2026). "CogAdapt: Adapting Clinical ECG Foundation Models for Wearable Cognitive Load Assessment". *Accepted at the IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI 2026)*. <https://arxiv.org/abs/2605.22774>.

4. Mousavi, A., **Sirjani, M. S.**, Nourbakhsh, E., Xie, M., Slavin, R., Neely, L., Davis, J., & Quarles, J. (2026). "MambaGaze: Bidirectional Mamba with Explicit Missing Data Modeling for Cognitive Load Assessment from Eye-Gaze Tracking Data". *Under review at the IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI 2026)*. <https://arxiv.org/abs/2605.22775>.
5. Nourbakhsh, E., **Sirjani, M. S.**, Mousavi, A., Nguyen, K., Quarles, J., Xie, M., & Slavin, R. (2026). "Are LLM Benchmarks Already Contaminated? A Systematic Review of Contamination Detection Methods". In Proceedings of the *Fifth Workshop on Generation, Evaluation and Metrics (GEM 2026)*, colocated with *ACL 2026*. **Outstanding Paper Award**. <https://doi.org/10.18653/v1/2026.gem-main.50>.
6. Ghaffari, A., Firoozi, V., Maleki, A., **Sirjani, M. S.**, & Abedini Bagha, M. (2025). "QTE-IoT: Q-Learning-Based Task Scheduling Scheme to Enhance Energy Consumption and QoS in IoT Environments". *Sustainable Computing: Informatics and Systems Journal*. <https://doi.org/10.1016/j.suscom.2025.101247>.
7. **Sirjani, M. S.**, Maleki, A., Pakmehr, A., Abedini Bagha, M., Ghaffari, A., & Pour Haji Kazem, A. A. (2025). "Controller placement in software-defined networks using reinforcement learning and metaheuristics". *Cluster Computing: The Journal of Networks, Software Tools and Applications*, 28(10), 660. <https://doi.org/10.1007/s10586-025-05331-y>.
8. **Sirjani, M. S.**, Ahmad, M., Mousavi, A., Nourbakhsh, E., & Nguyen, K. (2026). "Optimizing Task Scheduling in Fog Computing with Deadline Awareness". In Proceedings of the *2nd IEEE International Conference on Secure IoT, Assured and Trusted Computing (SATC 2026)*. IEEE. <https://doi.org/10.1109/SATC69565.2026.11542230>.
9. **Sirjani, M. S.**, Mousavi, S. A., & Sadeghi, M. (2024). "Data Mining and Cloud Computing for Customer Pattern Analysis and Value Maximization". In Proceedings of the *IEEE 10th International Conference on Artificial Intelligence and Robotics (QICAR 2024)* (pp. 339-344). IEEE. <https://doi.org/10.1109/QICAR61538.2024.10496623>.
10. Mousavi, S. A., **Sirjani, M. S.**, Bozorg Zadeh Razavi, S. J., & Nikooghadam, M. (2023). "SecVanet: provably secure authentication protocol for sending emergency events in VANET". In Proceedings of the *IEEE 14th International Conference on Information and Knowledge Technology (IKT 2023)* (pp. 86-91). IEEE. <https://doi.org/10.1109/IKT62039.2023.10433027>.
11. Mousavi, S. A., Sadeghi, M., & **Sirjani, M. S.** (2023). "A Comparative Evaluation of Machine Learning Algorithms for IDS in IoT network". In Proceedings of the *IEEE 14th International Conference on Information and Knowledge Technology (IKT 2023)* (pp. 168-174). IEEE. <https://doi.org/10.1109/IKT62039.2023.10433047>.

Awards & Honors

- Certificate, Building Agentic AI Applications with Large Language Models – NVIDIA, hosted by UTSA College of AI, Cyber and Computing, 2026
- Winner, 2-Minute Presentation Award Competition – 63rd Design Automation Conference (DAC), 2026
- Selected as a DAC Young Fellow – 63rd Design Automation Conference (DAC), 2026
- Graduate School Academic Travel Fund – University of Texas at San Antonio, 2026
- Outstanding Paper Award – Fifth Workshop on Generation, Evaluation and Metrics (GEM 2026), colocated with ACL 2026
- Fan Favorite Award – Draper Data Science Business Plan Competition, 2026
- Scholarship recipient – DAIR3 Data Science Summer School, 2026
- Winner, 2-Minute Presentation Award Competition – 62nd Design Automation Conference (DAC), 2025
- Selected as a DAC Young Fellow – 62nd Design Automation Conference (DAC), 2025

- Graduate School Academic Travel Fund – University of Texas at San Antonio, 2025

Professional Service

- Reviewer – 36th Great Lakes Symposium on VLSI (GLSVLSI), 2026

Teaching Experience

- **Operating Systems** (Fall 2025) – Dr. Sam Silvestro, UTSA
- **Data Science** (Summer 2025) – Dr. Amin Sahba, UTSA
- **Computer Organization** (Summer 2025) – Dr. Subhasish Das, UTSA
- **Operating Systems** (Spring 2025) – Dr. Mimi Xie, UTSA
- **Fundamentals of Cloud Computing** (Spring 2024) – Dr. Somayeh Sobati-Moghadam, FUM
- **Principles of Compiler Design** (Fall 2023) – Dr. Haleh Amintoosi, FUM
- **Fundamentals of Data Mining** (Fall 2023) – Dr. Behshid Behkamal, FUM
- **Fundamentals of Cloud Computing** (Fall 2023) – Dr. Somayeh Sobati-Moghadam, FUM
- **Principles of Compiler Design** (Spring 2023) – Dr. Haleh Amintoosi, FUM
- **Principles of Web Development** (Spring 2023) – Dr. Somayeh Sobati-Moghadam, FUM
- **Principles of Web Development** (Fall 2022) – Dr. Somayeh Sobati-Moghadam, FUM

Technical Skills

- **Programming Languages:** Python, C, C++, C#
- **ML/DL Frameworks:** TensorFlow, PyTorch, Keras, scikit-learn
- **IoT Protocols:** MQTT
- **Embedded Platforms:** NVIDIA Jetson (Orin Nano, Orin NX, AGX Orin, Nano, Xavier), Raspberry Pi 5, Arduino, STM32 ARM Cortex M4, MSP430FR5994
- **Tools & Technologies:** Docker, Kubernetes, Git, CUDA